SSC MTS 2024 question paper with answers

28 October 2024 · Shift 1

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

By selling a TV set for ₹26,000, a seller earns 30% profit. If the production cost is increased by 10%, what should be the new selling price of a set so as to gain 15% profit?

(A) ₹25,300
(B) ₹28,300
(C) ₹26,300
(D) ₹30,300

Q.2

The average of five numbers is 38. If one number is excluded, their average is 36. Find the excluded number.

(A) 49
(B) 41
(C) 46
(D) 38

Q.4

The HCF of 1.75, 5.6 and 7 is:

(A) 0.7
(B) 0.07
(C) 3.5
(D) 0.35

Q.6

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 6% per year rate of interest for 4 years.

(A) 600
(B) 540
(C) 640
(D) 560

Q.7

Tannish travels from A to B at a speed of 20 km/h, from B to C at 15 km/h and from C to D at 30 km/h. If AB = BC = CD, then find Tannish's average speed for the whole journey.

(A) 22 km/h
(B) 19 km/h
(C) 20 km/h
(D) 21 km/h

Q.8

A shopkeeper marks his good 12% above the cost price and allows a discount of 8.5% on the marked price. Find his profit or loss percentage.

(A) Profit 2.48%
(B) Loss 2.48%
(C) Loss 4.65%
(D) Profit 4.65%

Q.10

A man riding his bike covers 960 m in 64 sec. What is his speed in km/hr?

(A) 64
(B) 15
(C) 54
(D) 50

Q.12

The average weight of 40 athletes and 5 trainers is 72 kg. When six more athletes of average weight 65 kg are included in the group, what is the average weight (in kg) of the group? (Rounded off to 2 decimal places)

(A) 72.25
(B) 71.18
(C) 65.21
(D) 69.37

Q.15

A trader sells an item at 25% profit. Had he bought it at 15% more and sold it for ₹147 more, he would have earned a profit of 30%. Find the cost price of the item.

(A) ₹600
(B) ₹525
(C) ₹650
(D) ₹450

Q.16

The area of a rhombus with diagonals of lengths 4 cm and 5 cm, respectively, is:

(A) 10 cm^2
(B) 40 cm^2
(C) 20 cm^2
(D) 4.5 cm^2

Q.17

Find the average of all odd numbers between 10 and 30.

(A) 40
(B) 20
(C) 10
(D) 30

Q.18

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 6% per year rate of interest for 6 years.

(A) 660
(B) 760
(C) 720
(D) 680

Q.19

A family of seven minors and some adults has an average wheat consumption of 11.10 kg. The average consumption for minors is 5.5 kg per person, and for adults, it is 16 kg per person. Find the number of adults in the family.

(A) 8
(B) 10
(C) 5
(D) 6

Q.20

The number of marbles in three bags is in the ratio of 2 : 3 : 4. If 12 marbles are added to each bag, then the ratio changed to 8 : 11 : 14. The total number of marbles in the beginning was:

(A) 200
(B) 126
(C) 162
(D) 216

Reasoning Ability and Problem Solving 16 questions

Q.1

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 30 – 10 ÷ 63 × 9 + 15 =?

(A) 85
(B) 87
(C) 80
(D) 90

Q.2

Which of the following interchanges of signs would make the given equation correct? 57 – 19 × 18 + 9 ÷ 45 = 18

(A) ÷ and –
(B) + and ÷
(C) + and ×
(D) ÷ and ×

Q.3

What should come in place of the question mark (?) in the given series? 226 231 236 241 246 ?

(A) 254
(B) 252
(C) 253
(D) 251

Q.4

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (11, 19, 8) (14, 29, 15)

(A) (7, 35, 23)
(B) (22, 25, 9)
(C) (16, 28, 12)
(D) (18, 21, 7)

Q.5

LRPN is related to SYWU in a certain way based on the English alphabetical order. In the same way, IOMK is related to PVTR. To which of the following is GMKI related, following the same logic?

(A) NTRP
(B) NTPR
(C) TNPR
(D) TNRP

Q.6

What should come in place of the question mark (?) in the given series based on the English alphabetical order? WSF, BUK ,GWP ,LYU, ?

(A) PAY
(B) RZY
(C) QAZ
(D) RBZ

Q.7

A, B, C, D, E, F and G are sitting around a circular table facing away from the centre (but not necessarily in the same order). Only 3 people are sitting between A and D when counted from the right of A. B is sitting 2nd to the left of D. F is sitting immediate to the right of C. E is not an immediate neighbour of D. What is the position of F with respect to G?

(A) 2nd to the left
(B) 2nd to the right
(C) 3rd to the right
(D) 3rd to the left

Q.8

In a certain code language, ‘paint the wall’ is written as ‘jh dw qp’ and ‘wall of honour’ is written as ‘fh ut dw’. How is ‘wall’ written in the given language?

(A) jh
(B) dw
(C) ut
(D) qp

Q.9

TCKS is related to WFNV in a certain way based on the English alphabetical order. In the same way, YHPX is related to BKSA. To which of the following is HPXG related, following the same logic?

(A) KSJA
(B) KSJL
(C) KSAJ
(D) SKKA

Q.10

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some glasses are cups. All cups are plates. Conclusions: (I) At least some plates are glasses. (II) All cups are glasses.

(A) Neither conclusion (I) nor (II) follows
(B) Both conclusions (I) and (II) follow
(C) Only conclusion (I) follows
(D) Only conclusion (II) follows

Q.14

In a certain code language, ‘ONLY’ is coded as ‘2648’ and ‘NICE’ is coded as ‘7534’.What is the code for ‘N’ in the given code language?

(A) 7
(B) 6
(C) 2
(D) 4

Q.15

In a certain code language, 'GOLF' is coded as '6428' and 'OPEN' is coded as '7653'. What is the code for 'O' in the given code language?

(A) 6
(B) 7
(C) 5
(D) 4

Q.16

Seven people R, S, T, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). Only two people are sitting between S and G when counted from the right of S. Only two people are sitting between E and T when counted from the left of E. R is sitting second to the right of G. T is not an immediate neighbour of R. D is not an immediate neighbour of G. Who is sitting to the immediate left and to the immediate right of F, respectively?

(A) T and G
(B) G and T
(C) G and S
(D) T and S

Q.18

11 is related to 45 following a certain logic. Following the same logic, 3 is related to 13. To which of the following is 6 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 26
(B) 28
(C) 23
(D) 25

Q.19

In a certain code language, ‘i know you’ is written as ‘mo rc tp’ and ‘do they know’ is written as ‘hk mo be’. How is ‘know’ written in the given code language?

(A) be
(B) mo
(C) rc
(D) tp

Q.20

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All cups are plates. Some plates are glasses. Conclusions: (I) Some cups are not glasses. (II) Some glasses are not plates.

(A) Both conclusions I and II follow.
(B) Only conclusion I follows.
(C) Only conclusion II follows.
(D) Neither conclusion I nor II follows.

General Awareness 25 questions

Q.1

Which part of a flower develops into a seed?

(A) Pistil
(B) Stamen
(C) Ovule
(D) Petal

Q.2

Which of the following is an alarming effect of thermal pollution in water bodies?

(A) The amount of dissolved oxygen decreased significantly
(B) Changes in turbidity
(C) The random changes in pH values
(D) Increasing amount of carbon dioxide

Q.3

Which of the following Khayal artists is the founding father of Lalit Kala Kendra of the University of Pune?

(A) Bhimsen Joshi
(B) Ravi Shankar
(C) Shiv Kumar Sharma
(D) Kishan Maharaj

Q.4

According to Census 2011 of India, which metropolitan city has the highest population?

(A) Mumbai
(B) Kolkata
(C) Chennai
(D) Delhi

Q.5

An Ummathat folk dancer from Karnataka _____ had been honoured with Padma Shri in 2023.

(A) Sharan Rani Backliwal
(B) Rani Machaiah
(C) Devika Rani
(D) Rani Karnaa Nayak

Q.6

Who among the following presides over the meeting of the Council of Ministers in a state?

(A) Chief Secretary of the state
(B) Chief Minister of the state
(C) Chairman of the Legislative Council
(D) Speaker of the Legislative Assembly

Q.7

Which of the following states is characterised by salt marshes known as 'Rann'?

(A) Gujarat
(B) Jharkhand
(C) Maharashtra
(D) Punjab

Q.8

Who among the following wrote a book called ‘Indica’ based on his travels and experiences in Mauryan India?

(A) Diodorus
(B) Megasthenes
(C) Pliny
(D) Strabo

Q.9

Who among the following was the founder of 'Bhoodan Yajna' in India?

(A) Vinoba Bhave
(B) Vallabhbhai Patel
(C) Mahatma Gandhi
(D) Subhas Chandra Bose

Q.10
In which year was the microfinance movement formally launched in India on a pilot basis?

(A) 1980
(B) 2000
(C) 1992
(D) 2010

Q.11
Which Article of the Indian Constitution states the following? ‘Directive Principles of State Policy are fundamental in the governance of the country and it shall be the duty of the state to apply these principles in making laws.’

(A) Article 37
(B) Article 30
(C) Article 40
(D) Article 35

Q.12
Which of the following novels is based on Malgudi, a fictional town in south India?

(A) My Days
(B) Sacred Games
(C) The Guide
(D) A Suitable Boy

Q.13
Who was sworn in as the Governor of Chhattisgarh in February 2023?

(A) Biswabhusan Harichandan
(B) Satyadev Narayan Arya
(C) Anusuiya Uikey
(D) Ramesh Bais

Q.14
Which red and buff sandstone is the tallest tower in India built in the thirteenth century?

(A) Shaheed Minar
(B) Qutub Minar
(C) Jhulta Minar
(D) Chand Minar

Q.15
In which of the following years did India NOT win an Olympic gold medal in hockey?

(A) 1948
(B) 1952
(C) 1960
(D) 1956

Q.16
In which of the following states is the Jawara dance performed on the occasion of a successful harvest?

(A) Punjab
(B) Haryana
(C) Gujarat
(D) Madhya Pradesh

Q.17
Which of the following is NOT a function of mesosomes?

(A) DNA replication and distribution to daughter cells
(B) Locomotion
(C) Increase the surface area of the plasma membrane and enzymatic content
(D) Respiration

Q.18
Madhavpur Fair in Gujarat is a _____ day long event to celebrate Lord Krishna's marriage with Rukmani.

(A) three
(B) four
(C) five
(D) two

Q.19
In which sport is the term 'Dolphin Kick' commonly used?

(A) Long Jump
(B) Swimming
(C) High Jump
(D) Javelin Throw

Q.20
Who among the following founded a suburban township near Vijayanagara, called Nagalapuram, after his mother?

(A) Sadasiva Raya
(B) Achyuta Deva Raya
(C) Krishna Deva Raya
(D) Rama Raya

Q.21
According to National Multidimensional Poverty Index 2023 (A Progress Review), what was the proportion of population in multidimensional poverty (Headcount Ratio) in rural India in 2019-2021?

(A) 14.96%
(B) 12.32%
(C) 24.12%
(D) 19.28%

Q.22
Who among the following was a leader of Garam Dal, a group of radical nationalists?

(A) Gopal Krishna Gokhale
(B) Bipin Chandra Pal
(C) Dadabhai Naoroji
(D) Surendranath Banerjee

Q.23
The people of Himachal Pradesh make channels to bring water from a glacier or rainwater to villages for the function of mainly irrigation. These channels are locally known as:

(A) Baori
(B) Guls or Kuls
(C) Khadins
(D) Johads

Q.24
In December 2023, which eminent agricultural economist from Kerala was conferred the distinguished title of ISAE Fellow 2023 by the Indian Society of Agricultural Economics (ISAE)?

(A) Suman Sahai
(B) Dr. P Indira Devi
(C) Sulochana Gadgil
(D) Chanda Nimbkar

Q.25
Which of the following is NOT a hydroelectric power plant?

(A) Sambhar Lake Hydel Project
(B) Damodar Valley Hydel Project
(C) Kopili Hydel Project
(D) Bhakra Nangal Hydel Project

English Language and Comprehension 23 questions

Q.1
In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. Ahuja is a “cleverest” lawyer.

(A) clever
(B) more clever
(C) most clever
(D) cleverer

Q.2
Select the most appropriate synonym of the word 'Ceased' to fill in the blank. He was _____ from his job because of his misconduct.

(A) terminated
(B) progressed
(C) continued
(D) commenced

Q.3

Select the most appropriate ANTONYM of the underlined word in the given sentence. A few students have “condemned” the latest project of preparing teaching aids as a waste of money.

(A) Consolidated
(B) Contradicted
(C) Applauded
(D) Doomed

Q.5

Select the most appropriate option to fill in the blank. The discussion led by the scientist was very _____.

(A) most informative
(B) informative
(C) more informative
(D) information

Q.6

Select the most appropriate ANTONYM of the given word. Curse

(A) Abuse
(B) Praise
(C) Comment
(D) Criticise

Q.7

Select the most appropriate option that can substitute the underlined segment in the given sentence. Wolves’ teeth are larger and “strong” than dog’s teeth.

(A) the stronger
(B) stronger
(C) strongest
(D) more strong

Q.8

Select the most appropriate meaning of the given idiom. A blue-eyed boy

(A) an unwelcome intruder
(B) a miser
(C) an unperceived observer
(D) one who is favourite

Q.9

Select the most appropriate option to complete the given sentence in the present perfect continuous tense. The company _____ working towards achievements in curbing slackness and fostering development.

(A) is being
(B) are being
(C) has been
(D) have been

Q.10

Select the most appropriate synonym of the word 'Native' from the given sentence. Mr. Smith believes that various tribes are the indigenous habitants of various corners of the world; unlike his contemporaries who have a viewpoint regarding an alien or foreign invasion.

(A) Indigenous
(B) Foreign
(C) Alien
(D) Invasion

Q.12

Select the most appropriate synonym of the given word. Anticipate

(A) doubt
(B) Accuse
(C) Predict
(D) Conceal

Q.13

Select the most appropriate adjective to fill in the blank. She always wears dresses of _____ fashion.

(A) the latter
(B) the latest
(C) latest
(D) late

Q.14

The following sentence has been split into segments. Identify the segment that contains a grammatical error. If you don't find any error, mark 'No error' as your answer. He missed the college bus / because he woke up / unpardonably lately in the morning.

(A) because he woke up
(B) unpardonably lately in the morning
(C) no error
(D) he missed the college bus

Q.15

Select the most appropriate meaning of the given idiom. Crocodile tears

(A) An insincere display of grief
(B) To please everyone
(C) To not behave in a fair way
(D) An irrevocable step

Q.16

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in the blanks. Michael Jackson is known as the King of Pop. He _____ his own genre of music, Pop. His music is spreading its influence and popularity _____ the world.

(A) dominates; around
(B) presides; round
(C) command; besides
(D) controls; beside

Q.17

Select the correct spelling of the underlined word in the given sentence. The magician performed a "magnificant" trick.

(A) magnificante
(B) magneficent
(C) magnificent
(D) magnificient

Q.18

Select the most appropriate option to substitute the underlined segment in the given sentence. If no substitution is required, select 'No substitution required'. Manisha had a great feeling for those who were "junior than" her.

(A) junior to
(B) No substitution required
(C) junior of
(D) junior from

Q.19

Select the option that correctly rectifies the underlined spelling error in the given sentence. The reorganisation of daily life wrought by industrialisation had effects that weakened the material basis for the "instietutions" of the family and the community.

(A) instetuitions
(B) instetutions
(C) institutions
(D) instetusion

Q.20

Select the most appropriate option to fill in the blank. The sharpshooter _____ dead a man-eating tiger after it allegedly killed nine people in the last few days.

(A) killed
(B) threw
(C) fixed
(D) shot

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Worldwide illiteracy rates have (1)_____ declined in the last few decades. One of the major reasons for this decline is the sharp increase in literacy rates (2)_____ young women, which is a result of specific (3)_____ designed to increase educational opportunities for girls. (4)_____, there are still an estimated 771 million illiterate adults in the world, about two-thirds of whom are (5)_____.

Select the most appropriate option to fill in blank no. 1.

(A) thirdly
(B) consistently
(C) wearily
(D) consequently

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Worldwide illiteracy rates have (1)_____ declined in the last few decades. One of the major reasons for this decline is the sharp increase in literacy rates (2)_____ young women, which is a result of specific (3)_____ designed to increase educational opportunities for girls. (4)_____, there are still an estimated 771 million illiterate adults in the world, about two-thirds of whom are (5)_____.

Select the most appropriate option to fill in blank no. 2.

(A) between
(B) among
(C) over
(D) against

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Worldwide illiteracy rates have (1)_____ declined in the last few decades. One of the major reasons for this decline is the sharp increase in literacy rates (2)_____ young women, which is a result of specific (3)_____ designed to increase educational opportunities for girls. (4)_____, there are still an estimated 771 million illiterate adults in the world, about two-thirds of whom are (5)_____.

Select the most appropriate option to fill in blank no. 3.

(A) companions
(B) campaigns
(C) champions
(D) camisole

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Worldwide illiteracy rates have (1)_____ declined in the last few decades. One of the major reasons for this decline is the sharp increase in literacy rates (2)_____ young women, which is a result of specific (3)_____ designed to increase educational opportunities for girls. (4)_____, there are still an estimated 771 million illiterate adults in the world, about two-thirds of whom are (5)_____.

Select the most appropriate option to fill in blank no. 4.

(A) Whenever
(B) Otherwise
(C) However
(D) Thus

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Worldwide illiteracy rates have (1)_____ declined in the last few decades. One of the major reasons for this decline is the sharp increase in literacy rates (2)_____ young women, which is a result of specific (3)_____ designed to increase educational opportunities for girls. (4)_____, there are still an estimated 771 million illiterate adults in the world, about two-thirds of whom are (5)_____.

Select the most appropriate option to fill in blank no. 5.

(A) woman

(B) child

(C) man

(D) women

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.2 **(C)**	Q.4 **(D)**	Q.6 **(A)**	Q.7 **(C)**	Q.8 **(A)**
Q.10 **(C)**	Q.12 **(B)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(B)**	Q.18 **(C)**
Q.19 **(A)**	Q.20 **(C)**				

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(B)**	Q.9 **(C)**	Q.10 **(C)**	Q.14 **(D)**	Q.15 **(A)**
Q.16 **(A)**	Q.18 **(D)**	Q.19 **(B)**	Q.20 **(D)**		

General Awareness

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(A)**	Q.8 **(B)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(A)**	Q.12 **(C)**
Q.13 **(A)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(D)**	Q.17 **(B)**	Q.18 **(C)**
Q.19 **(B)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(B)**	Q.23 **(B)**	Q.24 **(B)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(C)**	Q.5 **(B)**	Q.6 **(B)**	Q.7 **(B)**
Q.8 **(D)**	Q.9 **(C)**	Q.10 **(A)**	Q.12 **(C)**	Q.13 **(B)**	Q.14 **(B)**
Q.15 **(A)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(A)**	Q.19 **(C)**	Q.20 **(D)**
Q.21 **(B)**	Q.22 **(B)**	Q.23 **(B)**	Q.24 **(C)**	Q.25 **(D)**	

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) ₹25,300

Cost = 26,000 ÷ 1.3 = ₹20,000. After a 10% rise it is ₹22,000, and a 15% profit on that gives 22,000 × 1.15 = ₹25,300.

Q.2 · Answer (C) 46

The five numbers total 5 × 38 = 190 and the remaining four 4 × 36 = 144, so the excluded number is 190 − 144 = 46.

Q.4 · Answer (D) 0.35

In hundredths the numbers are 175, 560 and 700: $175 = 5^2 \times 7$, $560 = 2^4 \times 5 \times 7$ and $700 = 2^2 \times 5^2 \times 7$, so their HCF is 35. The HCF of the decimals is 0.35.

Q.6 · Answer (A) 600

SI = 2,500 × 6 × 4/100 = ₹600.

Q.7 · Answer (C) 20 km/h

For three equal distances the average speed is 3 ÷ (1/20 + 1/15 + 1/30) = 3 ÷ (9/60) = 20 km/h.

Q.8 · Answer (A) Profit 2.48%

Selling price = 1.12 × 0.915 = 1.0248 times the cost price: a profit of 2.48%.

Q.10 · Answer (C) 54

960 m in 64 s is 15 m/s, and 15 × 18/5 = 54 km/h.

Q.12 · Answer (B) 71.18

The 45 people weigh 45 × 72 = 3,240 kg and the six athletes 6 × 65 = 390 kg, so the new average is (3,240 + 390) ÷ 51 = 3,630 ÷ 51 = 71.18 kg.

Q.15 · Answer (A) ₹600

With cost price x the selling price is 1.25x. At the new cost 1.15x a 30% profit needs 1.3 × 1.15x = 1.495x, so 1.495x − 1.25x = 0.245x = ₹147 and x = ₹600.

Q.16 · Answer (A) 10 cm²

Area of a rhombus = ½ × d_1 × d_2 = ½ × 4 × 5 = 10 cm².

Q.17 · Answer (B) 20

The odd numbers 11, 13, …, 29 are evenly spaced, so their average is (11 + 29)/2 = 20.

Q.18 · Answer (C) 720

SI = 2,000 × 6 × 6/100 = ₹720.

Q.19 · Answer (A) 8

With n adults: 7 × 5.5 + 16n = 11.10(7 + n), so 38.5 + 16n = 77.7 + 11.1n, 4.9n = 39.2 and n = 8.

Q.20 · Answer (C) 162

With 2x, 3x and 4x marbles: (2x + 12)/(3x + 12) = 8/11 gives 22x + 132 = 24x + 96, so x = 18 (48, 66 and 84 afterwards, which is 8 : 11 : 14). The total at first was 9 × 18 = 162.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) 85

After interchanging the signs: 30 + 10 × 63 ÷ 9 − 15 = 30 + 70 − 15 = 85.

Q.2 · Answer (A) ÷ and −

Interchanging ÷ and −: 57 ÷ 19 × 18 + 9 − 45 = 3 × 18 + 9 − 45 = 54 + 9 − 45 = 18.

Q.3 · Answer (D) 251

Each term is 5 more than the one before: 246 + 5 = 251.

Q.4 · Answer (C) (16, 28, 12)
The middle number is the sum of the other two: 11 + 8 = 19 and 14 + 15 = 29. Likewise 16 + 12 = 28.

Q.5 · Answer (A) NTRP
Each letter moves 7 places forward: L → S, R → Y, P → W, N → U, and I → P, O → V, M → T, K → R. So GMKI → NTRP.

Q.6 · Answer (C) QAZ
First letters move 5 forward (W, B, G, L, Q, wrapping round after Z), second letters 2 forward (S, U, W, Y, A) and third letters 5 forward (F, K, P, U, Z): QAZ.

Q.7 · Answer (C) 3rd to the right
Facing away from the centre, a person's right is clockwise. Clockwise from A the seats are A, E, B, G, D, C, F: D has three people between it and A, B is second to D's left, C and F take the only two adjacent free seats, and E, who cannot sit beside D, sits next to A. Counting clockwise from G, F is third: 3rd to the right.

Q.8 · Answer (B) dw
'wall' is the only word in both sentences and 'dw' the only code in both, so wall = dw.

Q.9 · Answer (C) KSAJ
Each letter moves 3 places forward (X wraps round to A): T → W, C → F, K → N, S → V. So HPXG → KSAJ.

Q.10 · Answer (C) Only conclusion (I) follows
Some glasses are cups and all cups are plates, so those glasses are plates: at least some plates are glasses (I). Nothing says every cup is a glass, so II does not follow.

Q.14 · Answer (D) 4
N is the only letter in both words and 4 the only digit in both codes, so N = 4.

Q.15 · Answer (A) 6
O is the only letter in both words and 6 the only digit in both codes, so O = 6.

Q.16 · Answer (A) T and G
Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise from S the seats are S, D, R, E, G, F, T, the only order that meets every condition (the other place for E and T puts T next to R). F has T on the left and G on the right.

Q.18 · Answer (D) 25
The second number is 4 times the first plus 1: 4 × 11 + 1 = 45 and 4 × 3 + 1 = 13. Likewise 4 × 6 + 1 = 25.

Q.19 · Answer (B) mo
'know' is the only word in both sentences and 'mo' the only code in both, so know = mo.

Q.20 · Answer (D) Neither conclusion I nor II follows.
Every cup could also be a glass, so I need not follow; and nothing places any glass outside the plates, so II need not follow either.

General Awareness

Q.1 · Answer (C) Ovule
After fertilisation the ovule develops into the seed and the ovary into the fruit.

Q.2 · Answer (A) The amount of dissolved oxygen decreased significantly
Warmer water holds less dissolved oxygen, so heated discharges can leave too little for fish and other aquatic life.

Q.3 · Answer (A) Bhimsen Joshi
Savitribai Phule Pune University credits Pandit Bhimsen Joshi, the Kirana gharana Khayal singer, as the founding father of its Lalit Kala Kendra (Centre for Performing Arts). Ravi Shankar, Shivkumar Sharma and Kishan Maharaj were instrumentalists (sitar, santoor and tabla).

Q.4 · Answer (A) Mumbai
In Census 2011 Greater Mumbai was the most populous urban agglomeration, about 1.84 crore, ahead of Delhi (about 1.63 crore), Kolkata (about 1.41 crore) and Chennai.

Q.5 · Answer (B) Rani Machaiah
Rani Machaiah, a Kodava folk dancer from Kodagu, received the Padma Shri in 2023 for the Ummathat dance.

Q.6 · Answer (B) Chief Minister of the state
The Chief Minister heads the state Council of Ministers (Article 163) and presides over its meetings.

Q.7 · Answer (A) Gujarat
The Rann of Kutch, a vast seasonal salt marsh, is in Gujarat.

Q.8 · Answer (B) Megasthenes
Megasthenes, Seleucus Nicator's envoy at the court of Chandragupta Maurya, wrote 'Indica'.

Q.9 · Answer (A) Vinoba Bhave
Vinoba Bhave began the Bhoodan (land gift) movement in 1951 at Pochampally in present-day Telangana.

Q.10 · Answer (C) 1992
NABARD launched the pilot of its Self-Help Group-Bank Linkage Programme in 1992, the formal start of microfinance in India.

Q.11 · Answer (A) Article 37
Article 37 makes the Directive Principles unenforceable by any court but fundamental in governance, and makes it the State's duty to apply them in making laws.

Q.12 · Answer (C) The Guide
R. K. Narayan's 'The Guide' is set in Malgudi. 'My Days' is his memoir; 'Sacred Games' is by Vikram Chandra and 'A Suitable Boy' by Vikram Seth.

Q.13 · Answer (A) Biswabhusan Harichandan
Biswabhusan Harichandan, until then Governor of Andhra Pradesh, was sworn in as Governor of Chhattisgarh in February 2023, succeeding Anusuiya Uikey.

Q.14 · Answer (B) Qutub Minar
The Qutub Minar in Delhi, of red and buff sandstone, was begun by Qutb-ud-din Aibak and completed by Iltutmish in the thirteenth century.

Q.15 · Answer (C) 1960
India won hockey gold in 1948, 1952 and 1956, but lost the 1960 final in Rome to Pakistan and took silver.

Q.16 · Answer (D) Madhya Pradesh
Jawara is a harvest dance of Bundelkhand in Madhya Pradesh, in which women dance balancing baskets of grain on their heads.

Q.17 · Answer (B) Locomotion
NCERT lists the mesosome's roles as cell wall formation, DNA replication and distribution to daughter cells, respiration, secretion, and increasing the plasma membrane's surface area and enzymatic content. Bacteria move with flagella.

Q.18 · Answer (C) five
The Madhavpur Ghed fair in Porbandar district runs for five days from Ram Navami, celebrating Krishna's marriage to Rukmini.

Q.19 · Answer (B) Swimming
The dolphin kick, with both legs moving up and down together, is the kick of butterfly swimming.

Q.20 · Answer (C) Krishna Deva Raya
Krishnadeva Raya built the township of Nagalapuram near Vijayanagara and named it after his mother.

Q.21 · Answer (D) 19.28%
NITI Aayog's National MPI 2023 puts rural India's headcount ratio at 19.28% in 2019-21, down from 32.59% in 2015-16; 14.96% is the figure for India as a whole.

Q.22 · Answer (B) Bipin Chandra Pal
Bipin Chandra Pal, with Bal Gangadhar Tilak and Lala Lajpat Rai (Lal-Bal-Pal), led the radical Garam Dal; Gokhale, Naoroji and Surendranath Banerjee were moderates.

Q.23 · Answer (B) Guls or Kuls
In the Western Himalayas, including Himachal Pradesh, people build diversion channels called 'guls' or 'kuls' to carry water to their fields.

Q.24 · Answer (B) Dr. P Indira Devi
Dr P. Indira Devi, an agricultural economist at Kerala Agricultural University, was named an ISAE Fellow for 2023 by the Indian Society of Agricultural Economics.

Q.25 · Answer (A) Sambhar Lake Hydel Project
Sambhar Lake in Rajasthan is a salt lake with no hydel project; Damodar Valley, Kopili (Assam) and Bhakra Nangal all generate hydroelectricity.

English Language and Comprehension

Q.1 · Answer (A) clever
SSC's key gives 'clever': no comparison is made, so the positive degree fits, 'Ahuja is a clever lawyer.' ('A most clever lawyer', with 'most' meaning 'very', is also possible in formal English, but 'clever' is the plain correction.)

Q.2 · Answer (A) terminated
'Ceased' means stopped or ended; to be 'terminated' from a job is to have it ended.

Q.3 · Answer (C) Applauded
To 'condemn' is to criticise strongly; its opposite is to 'applaud'.

Q.5 · Answer (B) informative
'Very' takes the positive degree: 'very informative'.

Q.6 · Answer (B) Praise
To 'curse' is to speak ill of or wish harm on; of the options, 'praise' is its opposite.

Q.7 · Answer (B) stronger
'Than' needs a comparative, matching 'larger': 'larger and stronger than'.

Q.8 · Answer (D) one who is favourite
A 'blue-eyed boy' is a favourite, someone treated with special favour.

Q.9 · Answer (C) has been
The present perfect continuous is 'has/have been' + -ing; 'the company' is singular, so 'has been working'.

Q.10 · Answer (A) Indigenous
'Native' means belonging to a place by birth or origin: 'indigenous'.

Q.12 · Answer (C) Predict
To 'anticipate' is to expect or foresee: to 'predict'.

Q.13 · Answer (B) the latest
'The latest fashion' is the newest one, and a superlative such as 'latest' takes 'the'; 'the latter' means the second of two.

Q.14 · Answer (B) unpardonably lately in the morning
'Lately' means recently; the adverb needed here is 'late': 'woke up unpardonably late'.

Q.15 · Answer (A) An insincere display of grief
'Crocodile tears' are a false show of sorrow.

Q.16 · Answer (A) dominates; around
He 'dominates' his genre, and its popularity spreads 'around' the world; 'presides' needs 'over', 'command' does not agree with 'he', and 'beside' means 'next to'.

Q.17 · Answer (C) magnificent
The correct spelling is 'magnificent'.

Q.18 · Answer (A) junior to
Latin comparatives such as 'junior' and 'senior' take 'to', not 'than': 'junior to her'.

Q.19 · Answer (C) institutions
The correct spelling is 'institutions'.

Q.20 · Answer (D) shot
'Shot dead' is the set phrase for killing with a gun.

Q.21 · Answer (B) consistently
Illiteracy rates have 'consistently', that is steadily, declined over the decades.

Q.22 · Answer (B) among
'Among' is used for a group: literacy rates 'among' young women.

Q.23 · Answer (B) campaigns
Specific 'campaigns' designed to increase educational opportunities for girls.

Q.24 · Answer (C) However
'However' introduces the contrast: despite the decline, 771 million adults are still illiterate.

Q.25 · Answer (D) women
About two-thirds of the illiterate adults are 'women': a plural, and the passage is about women's literacy.

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